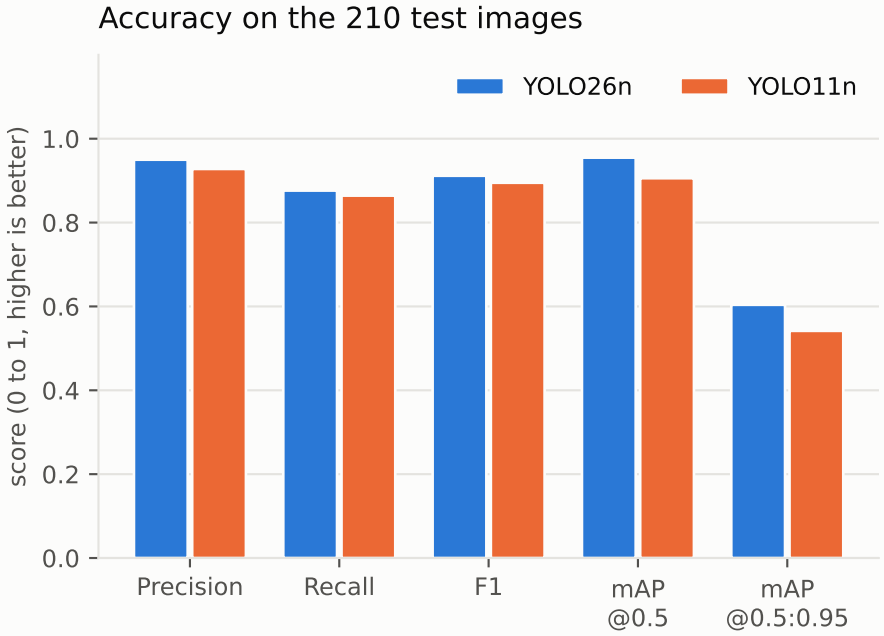
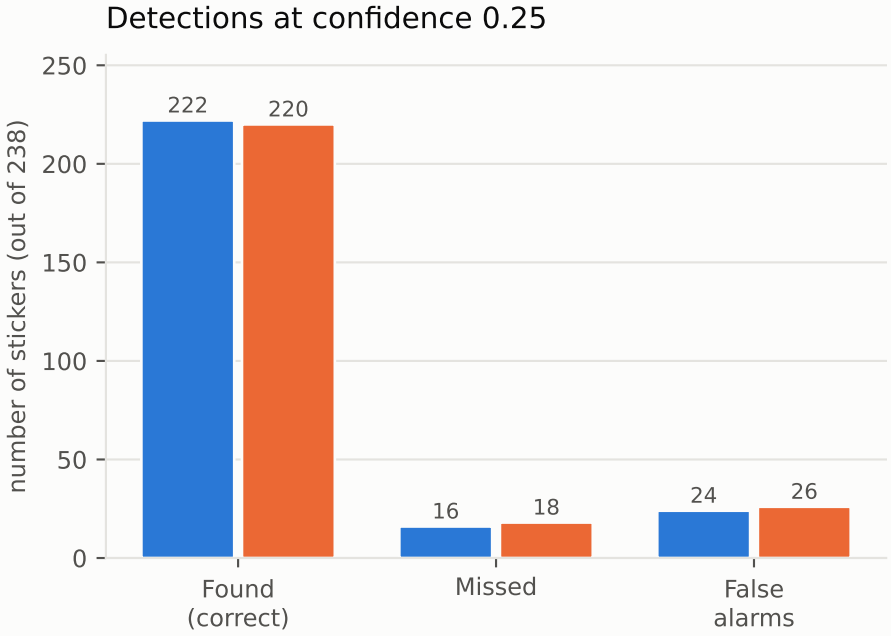
# YOLO26n vs YOLO11n

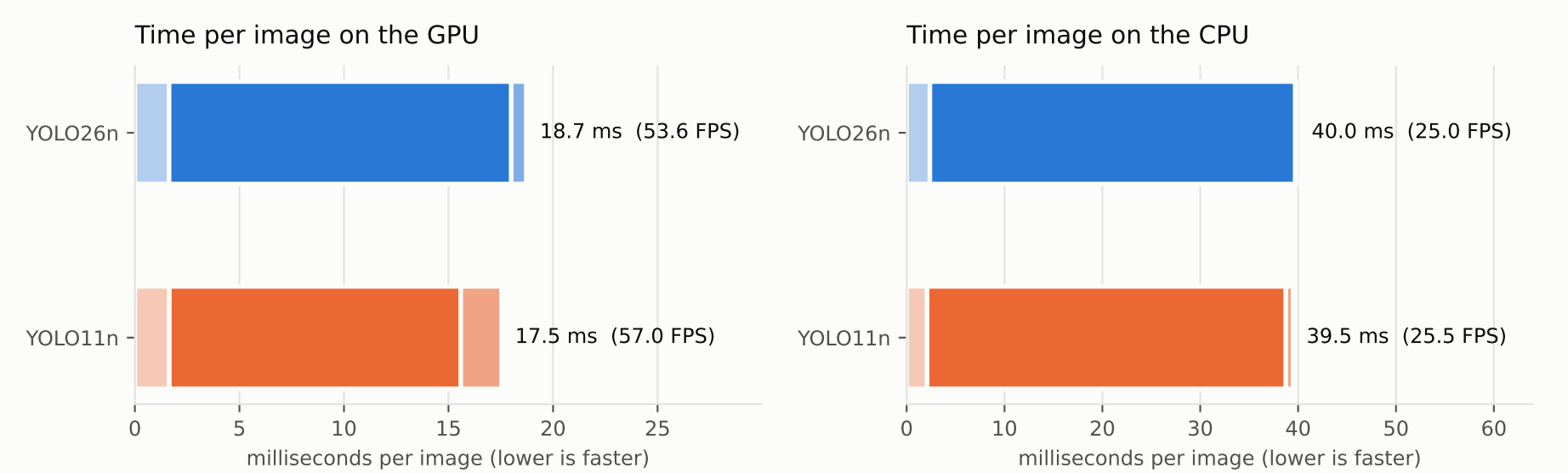
## PSU Vehicle Sticker Detection Benchmark

November 05, 2026

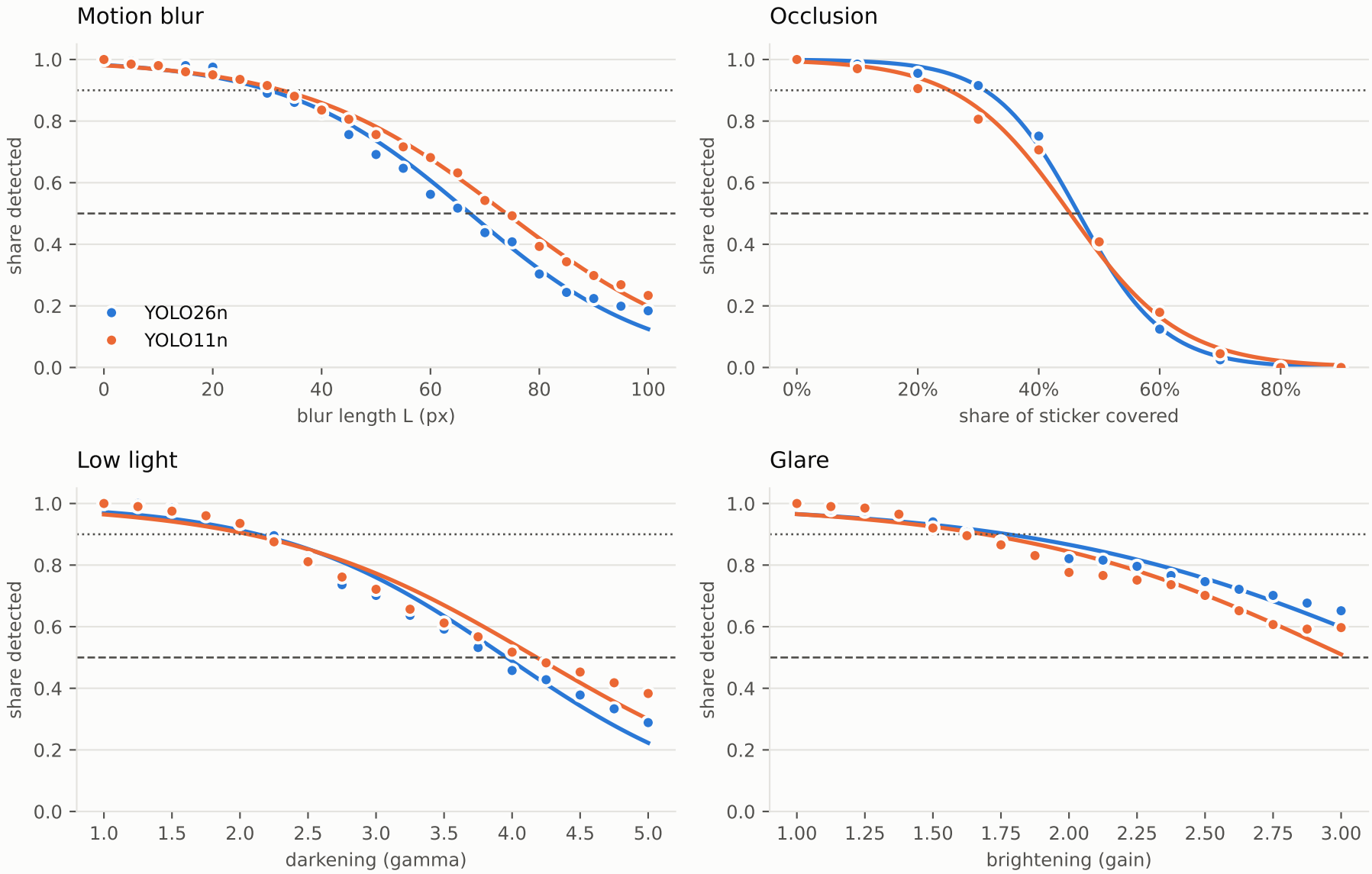
| Measure | YOLO26n | YOLO11n | Better |
|---|---|---|---|
| Accuracy: mAP@0.5:0.95 | 0.604 | 0.542 | YOLO26n |
| Accuracy: mAP@0.5 | 0.955 | 0.906 | YOLO26n |
| Precision | 0.950 | 0.928 | YOLO26n |
| Recall | 0.876 | 0.864 | YOLO26n |
| Speed on GPU (FPS) | 53.6 | 57.0 | About the same |
| Speed on CPU (FPS) | 25.0 | 25.5 | About the same |
| Motion blur: breaks at (50%) | 67 px | 74 px | YOLO11n |
| Occlusion: breaks at (50%) | 47% covered | 45% covered | No clear difference (YOLO26n slightly |
| Low light: breaks at (50%) | gamma 3.96 | gamma 4.18 | YOLO11n |
| Glare: breaks at (50%) | gain 3.27x | gain 3.02x | YOLO26n |


Accuracy on the 210 test images
YOLO26n
YOLO11n
score (0 to 1, higher is better)
1.0
0.8
0.6
0.4
0.2
0.0
Precision
Recall
F1
mAP
@0.5
mAP
@0.5:0.95

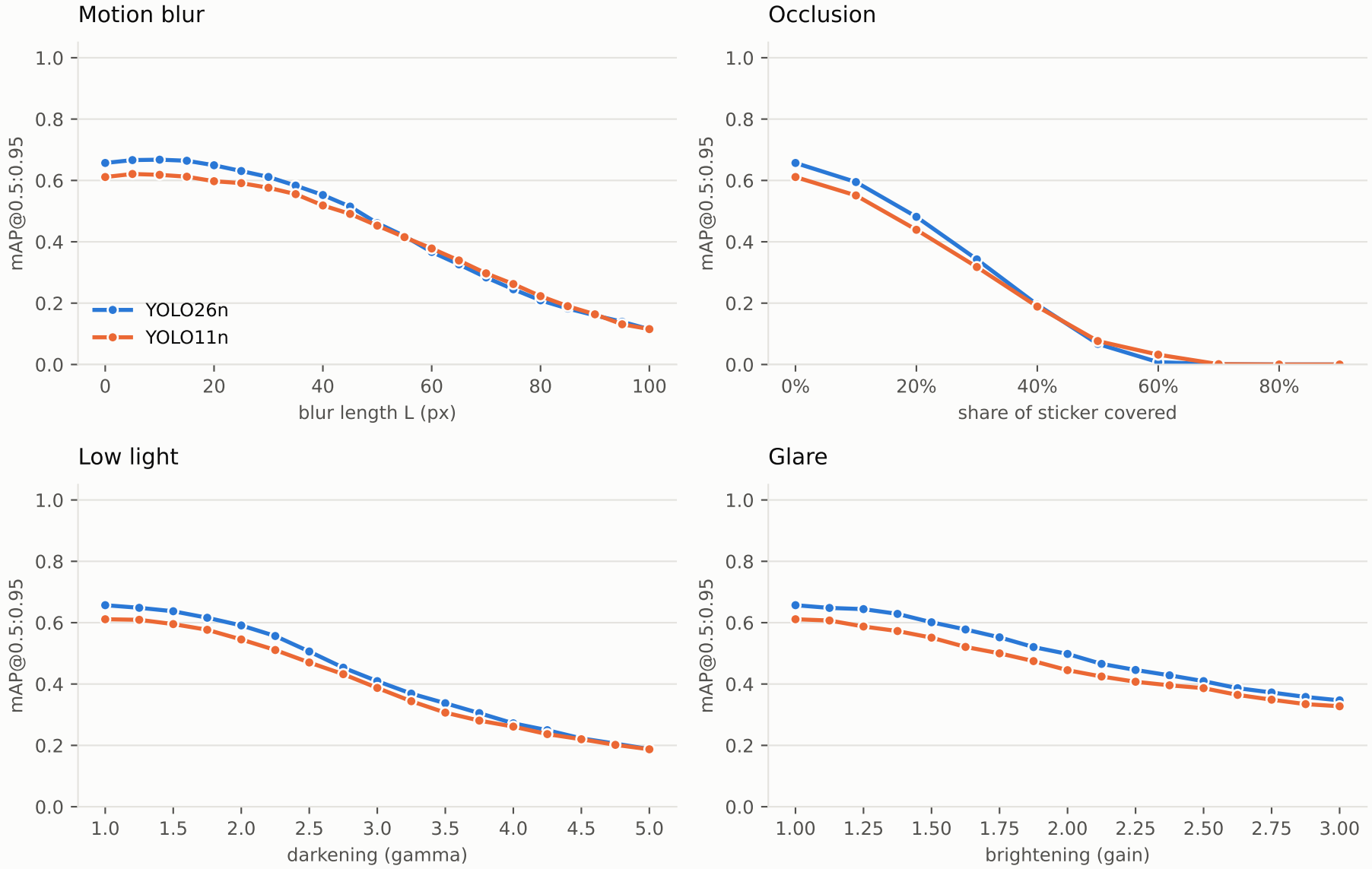

Detections at confidence 0.25
number of stickers (out of 238)
250
200
150
100
50
0
222
220
16
18
24
26
Found
(correct)
Missed
False
alarms



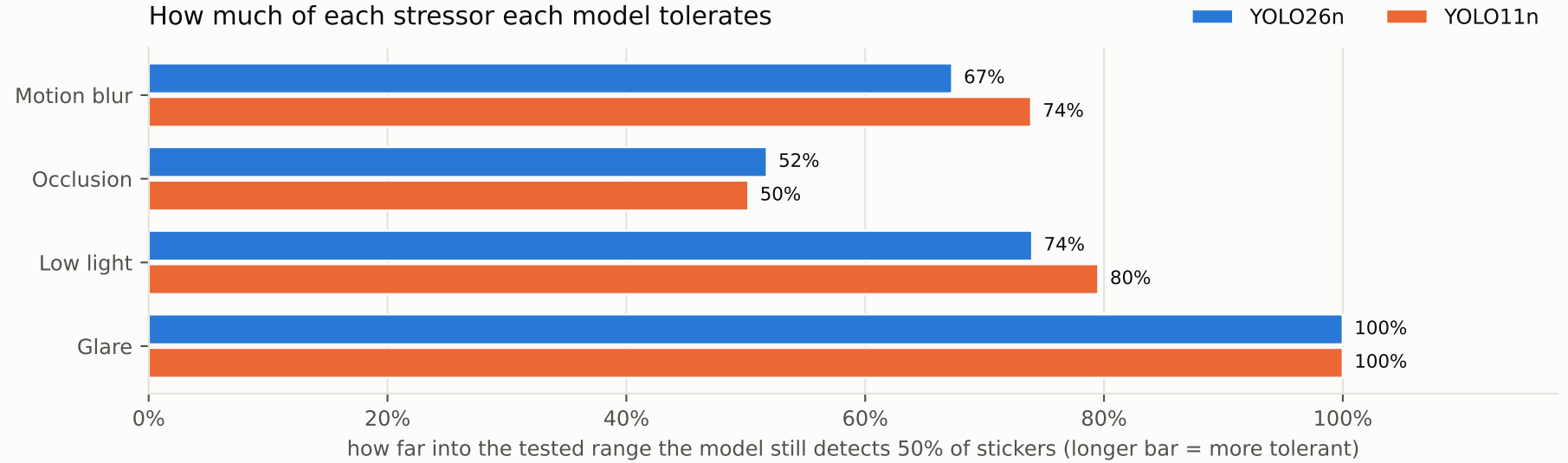

Bar shades, light to dark to medium: prepare image, model, cleanup (NMS). YOLO26n's cleanup is shorter because it is NMS-free.




Dots = measured, lines = fitted curve. Dashed line = breaking point (50% detected); dotted line = reliable detection (90%). Further right = harder.




Further right = harder.


How much of each stressor each model tolerates
YOLO26n
YOLO11n
Motion blur
67%
74%
Occlusion
52%
50%
Low light
74%
80%
Glare
100%
100%
0%
20%
40%
60%
80%
100%
how far into the tested range the model still detects 50% of stickers (longer bar = more tolerant)